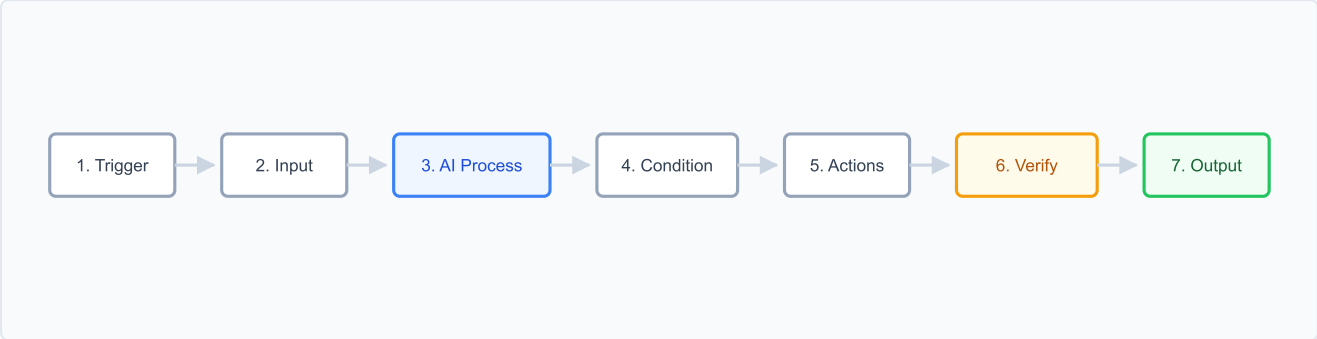
5. Standard AI Automation Formula

The Core Seven-Step Architecture for Intelligent Workflows

Building reliable, enterprise-grade automated systems requires structural discipline. Regardless of the specific tool or programming language used, robust AI implementations follow a standard seven-step formula. This progression ensures that intelligent models act strictly on clean data and are bound by deterministic logic.



Component Breakdown

Stage	Purpose and Function
1. Trigger	The event that initiates the workflow (e.g., an incoming webhook, a scheduled cron job, or a user clicking a submit button).
2. Input	Receiving, normalizing, and validating the initial payload before passing it downstream.
3. AI Processing	Using large language models strictly for analytical tasks: categorization, entity extraction, sentiment analysis, or summarization.
4. Conditions	Deterministic, hard-coded logic (if/else statements) that dictate how the payload is routed based on the AI's classification.
5. Actions	Executing the business logic, such as drafting a response, updating a database, or triggering an external API.
6. Verification	The safety net: checking outputs for compliance, or pausing the workflow to require a "Human-in-the-Loop" sign-off.
7. Output	The final resolution of the workflow—saving the terminal state and delivering the result to the end user or system of record.

Applied Example: Customer Support Routing

To understand how the formula acts as a blueprint, consider a common business requirement: automating the triage and initial response for incoming customer complaints.

> **The Scenario**
>
> *A customer submits a complaint → Validate the message → AI categorizes the issue → Apply routing rules → Prepare a response → Request approval where necessary → Save the result.*

Mapping the Scenario to the Formula

Step	Scenario Action	Technical Implementation Detail
1. Trigger	**Customer submits a complaint.**	A REST endpoint receives an HTTP POST request containing the ticket payload from the frontend web form.
2. Input	**Validate the message.**	The system ensures the payload contains a valid email and that the message body is not empty or malformed.
3. AI Process	**AI categorizes the issue.**	An LLM is prompted to classify the text into predefined labels (e.g., `Billing`, `Technical`, or `Refund`).
4. Conditions	**Apply routing rules.**	A switch statement or if/else block routes `Billing` to finance, and routes `Technical` to a specific support tier.
5. Actions	**Prepare a response.**	The system generates a drafted email acknowledging the specific issue using a predefined template.
6. Verification	**Request approval where necessary.**	If the AI classified the issue as `Refund`, the workflow pauses and alerts a human manager to approve the drafted email.
7. Output	**Save the result.**	The finalized ticket state, category, and action taken are committed to the PostgreSQL database for record-keeping.